

7.6 Classwork: Natural Logarithms and $\exp$ — Markscheme

1. [Maximum mark: 8]

(a) Evaluate.

(i) $\log_9 27$. Let $x = \log_9 27$, so $9^x = 27$, i.e. $3^{2x} = 3^3$. Hence $x = \frac{3}{2}$. **A1**

(ii) $\log_{27} 9 = \frac{2}{3}$ (since $27^{2/3} = (3^3)^{2/3} = 3^2 = 9$). **A1**

(iii) $\log_{1/3} 27 = -3$ (since $(1/3)^{-3} = 3^3 = 27$). **A1**

(iv) $\log_2 \sqrt[4]{8} = \log_2 (2^{3/4}) = \frac{3}{4}$. **A1**

(b) Let $L = \log_a b$.

(i) $\log_a (a^3 b) = \log_a a^3 + \log_a b = 3 + \log_a b$. **A1**

(ii) $\log_a \left(\frac{b^2}{a} \right) = 2 \log_a b - \log_a a = 2 \log_a b - 1$. **A1**

(iii) $\log_a \sqrt{a^3 b} = \frac{1}{2} \log_a (a^3 b) = \frac{1}{2} (3 + \log_a b) = \frac{3}{2} + \frac{1}{2} \log_a b$. **A1**

(iv) $\log_{a^3} b = \frac{\log_a b}{\log_a a^3} = \frac{1}{3} \log_a b$. **A1**

2. [Maximum mark: 5]

(a) $\ln(e^4) = 4.$ **A1**

(b) $e^{\ln 7} = 7.$ **A1**

(c) $\ln(e^{2x+1}) = 2x + 1.$ **A1**

(d) $e^{3 \ln 2} = e^{\ln 2^3} = e^{\ln 8}.$ **M1**

$= 8.$ **A1**

M1 may also be awarded for $e^{3 \ln 2} = (e^{\ln 2})^3 = 2^3.$

3. [Maximum mark: 6]

(a) $e^x = 12 \Rightarrow x = \ln 12.$ **A1**

(b) $\ln x = 5 \Rightarrow x = e^5.$ **A1**

(c) $e^{2x-1} = 9.$ Take $\ln$ of both sides: $2x - 1 = \ln 9.$ **M1**

$x = \frac{1 + \ln 9}{2}.$ **A1**

Accept $\frac{1}{2} + \ln 3$ (since $\ln 9 = 2 \ln 3$).

(d) $\ln(3x + 1) = 2.$ Exponentiate: $3x + 1 = e^2.$ **M1**

$x = \frac{e^2 - 1}{3}.$ **A1**

4. [Maximum mark: 13]

$$N(t) = 250 e^{kt}, \text{ with } N(3) = 1000.$$

(a) $N(0) = 250 e^0 = 250.$ **A1**

(b) Substitute $t = 3$ and $N = 1000$: $1000 = 250 e^{3k}.$ **M1**

$$e^{3k} = 4.$$
 A1

$$3k = \ln 4 \Rightarrow k = \frac{1}{3} \ln 4.$$
 A1 AG

(c) $N(5) = 250 e^{5k} = 250 \cdot 4^{5/3}.$ **M1**

$$4^{5/3} = 10.0794 \dots$$
 A1

$$N(5) = 250 \times 10.0794 \dots = 2519.84 \dots \approx 2520.$$
 A1

(d) Set $250 e^{kt} = 10\,000$, so $e^{kt} = 40.$ **M1**

$$kt = \ln 40 \Rightarrow t = \frac{\ln 40}{k} = \frac{3 \ln 40}{\ln 4}.$$
 M1

$$t = 7.9829 \dots \approx 7.98 \text{ hours (3 s.f.).}$$
 A1

M1 M1 may also be awarded for solving $250 e^{kt} = 10\,000$ with a graphing-calculator intersect or solver.

(e) Doubling: $2N_0 = N_0 e^{kt} \Rightarrow e^{kt} = 2.$ **M1**

$$t = \frac{\ln 2}{k} = \frac{\ln 2}{(1/3) \ln 4} = \frac{3 \ln 2}{2 \ln 2} = \frac{3}{2}.$$
 M1

$$t = 1.50 \text{ hours.}$$
 A1

5. [Maximum mark: 12]

$$A(t) = 5000 e^{0.06 t}.$$

$$\text{(a)} \quad A(4) = 5000 e^{0.24}. \quad \text{M1}$$

$$= 5000 \times 1.27124915 \dots = 6356.245 \dots \approx \$6356.25. \quad \text{A1}$$

$$\text{(b)} \quad 5000 e^{0.06 t} = 8000 \Rightarrow e^{0.06 t} = 1.6. \quad \text{M1}$$

$$0.06 t = \ln 1.6 \Rightarrow t = \frac{\ln 1.6}{0.06}. \quad \text{M1}$$

$$t = 7.8333 \dots \approx 7.83 \text{ years (3 s.f.)}. \quad \text{A1}$$

$$\text{(c)} \quad \text{Doubling: } 2 = e^{0.06 t}. \quad \text{M1}$$

$$t = \frac{\ln 2}{0.06}. \quad \text{M1}$$

$$t = 11.552 \dots \approx 11.6 \text{ years (3 s.f.)}. \quad \text{A1}$$

$$\text{(d)} \quad \text{Annual compounding: } A_{\text{ann}}(4) = 5000 (1.06)^4. \quad \text{M1}$$

$$= 5000 \times 1.26247696 = \$6312.38. \quad \text{A1}$$

$$\text{Difference: } \$6356.25 - \$6312.38 = \$43.87 \text{ (or } \$43.86 \text{ from unrounded subtraction).}$$

M1 A1

Accept \$43.86 or \$43.87.